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The Bunch–Kaufman factorization of symmetric matrices signature similar to sign regular matrices ☆

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ABSTRACT

The Bunch–Kaufman pivoting strategy is a most commonly used method in practice to factor symmetric indefinite matrices. However, this method in general may destroy the band structure of banded matrices, and its growth factor bound of $(2.57)^{n-1}$ can almost be attained. In this paper, we investigate the behavior of the method for factorizing symmetric indefinite matrices that are signature similar to sign regular matrices with signature ϵ. The growth factor bound depending on flips in the signature ϵ is derived. For factorizing such banded matrices, if the Bunch–Kaufman pivoting strategy is modified as suggested by Sorensen and Van Loan, then not only do the same growth factor bounds hold, but also the bandwidth is nicely preserved.

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1. Introduction

Let $\langle n\rangle = \{1, 2, \ldots, n\}$, and let $Q_{k,n}$ be the set of strictly increasing sequences of k natural numbers less than or equal to n. Denote by $A[\alpha|\beta]$ the submatrix of $A \in \mathbb{R}^{n\times n}$ with rows and columns indexed by $\alpha \in Q_{k,n}$ and $\beta \in Q_{k,n}$, respectively. The principal submatrix $A[\alpha|\alpha]$ is abbreviated to $A[\alpha]$. If $A[\alpha|\beta]$ is nonsingular, then the Schur complement $A/A[\alpha|\beta]$ is defined as follows:

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$$A/A[\alpha|\beta] = A[\alpha'|\beta'] - A[\alpha'|\beta](A[\alpha|\beta])^{-1}A[\alpha|\beta']$$

where $\alpha' = \langle n\rangle\backslash\alpha$ and $\beta' = \langle n\rangle\backslash\beta$. Denote by $I_n \in \mathbb{R}^{n\times n}$ the identity matrix. A vector $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$ is called a *signature sequence* when $|\epsilon_i| = 1$ for all i. The number of flips in a signature sequence $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$ $(n \geqslant 2)$ is defined by

$$\phi(\epsilon) = |\{i|\epsilon_i \neq \epsilon_{i+2},\ i = 0, 1, \ldots, n-2, \quad \text{where } \epsilon_0 = 1\}|.$$

A matrix $A \in \mathbb{R}^{n\times n}$ is *sign regular with signature* ϵ if $\epsilon_k \det A[\alpha|\beta] \geqslant 0$ for all $\alpha, \beta \in Q_{k,n}$ with any $k \in \langle n\rangle$. In particular, if all $\epsilon_k = 1$, then A is totally nonnegative; if all $\epsilon_k = -1$, then A is totally nonpositive. In the last years, sign regular matrices including totally nonnegative matrices and totally nonpositive matrices have been studied extensively [1,12,14,20,22] and have found a wide variety of applications [18,21]. For example, shape preserving representations in computer-aided geometric design are associated with sign regular matrices, and linear systems associated with variation-diminishing transformations have a close connection with sign regular linear systems [21]. Therefore, it is natural to consider how to factor sign regular matrices and solve sign regular linear systems with numerical stability. Indeed, some results have been obtained for totally nonnegative matrices [2] and sign regular matrices [20].

Our interest in the present paper is focused on how to factorize symmetric sign regular matrices by preserving the symmetry. Clearly, the first-last pivoting strategy in [20] is a stable method for symmetric sign regular matrices, however, this approach does not respect the symmetry. An alternative method is to construct a factorization LDL^T without pivoting, where L is lower triangular and D is diagonal, but this factorization may not exist. To take advantage of the symmetric properties, a commonly used method is a diagonal pivoting method which computes a factorization as follows: let $A \in \mathbb{R}^{n\times n}$ be symmetric, then one can choose a permutation matrix Π associated with a pivoting strategy such that

$$\Pi A \Pi^T = \begin{bmatrix} E & C^T \\ C & B \end{bmatrix} = \begin{bmatrix} I_s & 0 \\ CE^{-1} & I_{n-s} \end{bmatrix} \begin{bmatrix} E & 0 \\ 0 & B - CE^{-1}C^T \end{bmatrix} \begin{bmatrix} I_s & E^{-1}C^T \\ 0 & I_{n-s} \end{bmatrix},$$

where E is an $s \times s$ nonsingular pivot with $s = 1$ or 2. Repetition of this procedure on the Schur complement $S = B - CE^{-1}C^T$ then leads to a block LDL^T factorization $PAP^T = LDL^T$, where P is the associated permutation matrix, L is unit lower triangular and D is block diagonal with 1×1 and 2×2 blocks. Obviously, the pivoting strategy here is crucial for achieving numerical stability. There are three well-known pivoting strategies in the literature: the Bunch–Parlett pivoting strategy, the Bunch–Kaufman pivoting strategy and the symmetric rook pivoting strategy. Among these pivoting strategies, the Bunch–Kaufman pivoting strategy is the most widely used method, which has been chosen for both LAPACK and LINPACK. Now let us recall the pivoting strategy in Algorithm 1 by referring to [8, Algorithm 1], where we denote the permutation matrix

$$Q_r = \begin{bmatrix} 0 & 1 & 0 \\ I_{r-1} & 0 & 0 \\ 0 & 0 & I_{n-r} \end{bmatrix} \in \mathbb{R}^{n\times n}, \quad \forall\ 1 \leqslant r \leqslant n.$$

In addition, Sorensen and Van Loan [5, Section 5.3.2] proposed to modify the Bunch–Kaufman pivoting strategy by replacing $\sigma =: \max_{i\neq r} |a_{ir}|$ with $\sigma =: \max_i |a_{ir}|$. It must be pointed out that this small modification does not affect any of the main properties of this method such as the growth factor bound and the rounding error analysis. A thorough backward error analysis of the Bunch–Kaufman pivoting strategy has been provided by Higham [8] (also see [10, Chapter 11]). According to [8, Theorem 4.2], if the pivoting strategy is applied to solve a symmetric linear system $Ax = b$, then the computed solution $\hat{x}$ satisfies that

Algorithm 1

This algorithm determines the pivot for the first stage of the Bunch–Kaufman pivoting strategy applied to a symmetric matrix $A = (a_{ij}) \in \mathbb{R}^{n\times n}$.

$\alpha := (1+\sqrt{17})/8 (\approx 0.64)$
$\lambda =: \max_{i\neq 1} |a_{i1}|$
if $\lambda = 0$, then there is nothing to do on this stage of the factorization.
$r =: \min\{i \geqslant 2 : |a_{i1}| = \lambda\}$
if $|a_{11}| \geqslant \alpha\lambda$ **then**
  Use a_{11} as a 1×1 pivot (the associated permutation matrix $\Pi = I_n$).
else
  $\sigma =: \max_{i\neq r} |a_{ir}|$
  if $|a_{11}|\sigma \geqslant \alpha\lambda^2$ **then**
    Use a_{11} as a 1×1 pivot (the associated permutation matrix $\Pi = I_n$).
  else if $|a_{rr}| \geqslant \alpha\sigma$ **then**
    Use a_{rr} as a 1×1 pivot (the associated permutation matrix $\Pi = Q_r$).
  else
    Use $\begin{bmatrix} a_{11} & a_{1r} \\ a_{r1} & a_{rr} \end{bmatrix}$ as a 2×2 pivot (the associated permutation matrix $\Pi = \mathrm{diag}(1, Q_{r-1})$).
  end if
end if

$$(A+\triangle A)\hat{x} = b, \quad \|\triangle A\|_M \leqslant p(n)\rho_n\mu\|A\|_M + O(\mu^2), \tag{1}$$

where $p(n)$ is a quadratic, μ is the unit roundoff, the norm $\|\cdot\|_M$ is defined by $\|A\|_M = \max_{i,j}|a_{ij}|$, and the growth factor ρ_n is defined as

$$\rho_n = \frac{\max_{i,j,k}\left|a_{ij}^{(k)}\right|}{\max_{i,j}|a_{ij}|}$$

where the entries $a_{ij}^{(k)}$ are the elements that occur during the factorization. Obviously, the fact (1) means that the actual performance of this method is affected by the growth factor ρ_n. Large element growth usually leads to instability. It has been known for decades that the growth factor

$$\rho_n \leqslant \left(1+\frac{1}{\alpha}\right)^{n-1} \approx (2.57)^{n-1}.$$

However, as pointed out in [8, p. 63], there is even less known about the growth factor of this method than that of Gaussian elimination [7,11,16]. For many years, whether the growth factor bound of $(1+\frac{1}{\alpha})^{n-1}$ is attainable has remained an open question [10]. Recently, Druinsky and Toledo [4] have shown that the growth factor ρ_n can be as much as $\left(1+\frac{1}{\alpha}\right)^{n-2}$. Because the growth factor bound can almost be attained, it is natural to investigate classes of matrices where small growth factors are guaranteed, just as for Gaussian elimination. It is well known that the growth factor bound of 2^{n-1} for Gaussian elimination with partial pivoting can be attained, but for diagonally dominant matrices and their inverses [6], as well as for complex symmetric positive definite matrices [9], the growth factor is nicely bounded by 2.

Recently, the first author [15] has investigated some properties of the Bunch–Parlett pivoting strategy applied to special sign regular matrices, i.e., symmetric totally nonpositive matrices and their inverses, where the growth factor bound, the lower triangular factor bound and comparison costs are discussed. In particular, it has been shown that the growth factor of the pivoting strategy can be nicely bounded by 4 for these matrices.

Therefore, motivated by these results above, one aim of this paper is to identify classes of matrices which have highly desirable growth factor bounds for the Bunch–Kaufman pivoting strategy.

Additionally, it must be pointed out that the Bunch–Kaufman pivoting strategy in general does not preserve the band structure for banded matrices. In [3] it has been shown how to specialize the pivoting method for tridiagonal and 5 diagonal matrices. Jones and Patrick [17] described how to implement the Bunch–Kaufman factorization for banded real symmetric matrices. An algorithm has been designed in order to preserve the band structure [19].

A signature matrix is a diagonal matrix whose diagonal elements are 1 or -1. For a matrix $A \in \mathbb{R}^{n\times n}$, we say that A is signature similar to $B \in \mathbb{R}^{n\times n}$ if $B = SAS$ where $S \in \mathbb{R}^{n\times n}$ is a signature matrix. Different from the work in [15], our work in this paper is to investigate the behavior of the growth factor of the Bunch–Kaufman factorization of symmetric indefinite matrices that are signature similar to sign regular matrices with signature $\epsilon \in \mathbb{R}^{1\times n}$. We also investigate the effect of the factorization on the bandwidth of the matrix whenever it is banded. The rest of the paper is organized as follows. In Section 2, we show that for the Bunch–Kaufman factorization of such symmetric matrices, the growth factor ρ_n is bounded by $\left(1+\frac{1}{\alpha}\right)^{2\phi(\epsilon)} \approx (2.57)^{2\phi(\epsilon)}$. As a consequence, pleasantly small growth factor bounds of $(2.57)^2$ and 2.57 are obtained for symmetric totally nonpositive matrices and their inverses, respectively. In Section 3, we present that for the modified Bunch–Kaufman factorization of such banded symmetric matrices, not only do the same growth factor bounds as those of the Bunch–Kaufman factorization hold, but also the band structure is preserved.

2. The Bunch–Kaufman factorization

In this section, we present the growth factor bounds for the Bunch–Kaufman factorization of symmetric matrices that are signature similar to sign regular matrices.

The following result can be directly derived from [13, Theorem 1].

Lemma 1 [13]. *Let $A \in \mathbb{R}^{n\times n}$ be sign regular with signature $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$. If $A[\alpha]$ is nonsingular where $\alpha \in Q_{k,n}$, then there exists a signature matrix S such that*

$$S(A/A[\alpha])S$$

is sign regular with signature $\left(\frac{\epsilon_{k+1}}{\epsilon_k}, \frac{\epsilon_{k+2}}{\epsilon_k}, \ldots, \frac{\epsilon_n}{\epsilon_k}\right)$. In particular, if $\langle n\rangle\backslash\alpha$ is an index set consisting of consecutive integers, then $A/A[\alpha]$ is sign regular with signature $\left(\frac{\epsilon_{k+1}}{\epsilon_k}, \frac{\epsilon_{k+2}}{\epsilon_k}, \ldots, \frac{\epsilon_n}{\epsilon_k}\right)$.

All of the nonzero entries of a sign regular matrix $A \in \mathbb{R}^{n\times n}$ must have the same sign. Denote that sign by $\epsilon_1(A)$. Next we give the behavior of the element growth when one step of the Bunch–Kaufman factorization is performed.

Lemma 2. *Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ $(n \geqslant 2)$ be symmetric and signature similar to a nonsingular sign regular matrix with signature $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$, and let $A^{(1)}$ be the obtained Schur complement when one step of the Bunch–Kaufman factorization is applied to A. Then $A^{(1)} = \left(a_{ij}^{(1)}\right)$ is signature similar to a nonsingular sign regular matrix with signature $\left(\frac{\epsilon_{s+1}}{\epsilon_s}, \frac{\epsilon_{s+2}}{\epsilon_s}, \ldots, \frac{\epsilon_n}{\epsilon_s}\right)$, and for all i and j,*

$$\left|a_{ij}^{(1)}\right| \leqslant \begin{cases} \max\limits_{i,j} |a_{ij}|, & \text{if } \epsilon_2 = 1; \\ (2.57)^s \max\limits_{i,j} |a_{ij}|, & \text{if } \epsilon_2 = -1; \end{cases}$$

where $s = 1$ or 2 is the order of the pivot. In particular, if $\epsilon_2 = 1$, then $s = 1$.

Proof. Assume that there exists a signature matrix $S \in \mathbb{R}^{n\times n}$ such that

$$B = SAS = (b_{ij}) \tag{2}$$

is nonsingular sign regular with signature $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$. First let $\epsilon_2 = 1$. Then we have that

$$\det A[i,j] = \det B[i,j] = |a_{ii}||a_{jj}| - (a_{ij})^2 \geqslant 0,\ \forall\, i \neq j. \tag{3}$$

Now we choose the pivot according to Algorithm 1 when one step of the Bunch–Kaufman factorization is applied to A.

- Case 1. The pivot is a_{11}. Thus, the obtained Schur complement is

  $$A^{(1)} = A/A[1] = \left(a_{ij}^{(1)}\right).$$

  By using Lemma 1 and the fact (2), we have that

  $$B^{(1)} = B/B[1] = S'(A/A[1])S' = S'A^{(1)}S' = \left(b_{ij}^{(1)}\right) \tag{4}$$

  is nonsingular sign regular with signature $\left(\frac{\epsilon_2}{\epsilon_1}, \frac{\epsilon_3}{\epsilon_1}, \ldots, \frac{\epsilon_n}{\epsilon_1}\right)$, where S' is a signature matrix obtained from S by deleting its first row and column. In particular, $\epsilon_1(B^{(1)}) = \epsilon_1(B)$ by considering $\epsilon_2 = 1$. Hence, for all i and j,

  $$\left|a_{ij}^{(1)}\right| = |b_{ij}^{(1)}| = |b_{i+1,j+1} - \frac{b_{i+1,1}b_{1,j+1}}{b_{11}}| \leqslant |b_{i+1,j+1}| = |a_{i+1,j+1}| \leqslant \max_{i,j} |a_{ij}|.$$

- Case 2. The pivot is a_{rr}. Thus, the obtained Schur complement is

  $$A^{(1)} = A/A[r] = \left(a_{ij}^{(1)}\right).$$

  By Lemma 1 and the fact (2), there exists a signature matrix S_1 such that

  $$B^{(1)} = S_1(B/B[r])S_1 = S_1S'(A/A[r])S'S_1 = S_1S'A^{(1)}S'S_1 = \left(b_{ij}^{(1)}\right) \tag{5}$$

  is nonsingular sign regular with signature $(\frac{\epsilon_2}{\epsilon_1}, \frac{\epsilon_3}{\epsilon_1}, \ldots, \frac{\epsilon_n}{\epsilon_1})$, where S' is a signature matrix obtained from S by deleting its rth row and column. Since the pivot is a_{rr}, Algorithm 1 guarantees that

  $$|a_{11}| < \alpha\lambda, \quad |a_{11}|\sigma < \alpha\lambda^2,$$

  where

  $$\lambda = |a_{r1}| = \max_{i\neq 1} |a_{i1}|, \quad \sigma = \max_{i\neq r} |a_{ir}|,$$

  and thus, we must have $|a_{rr}| > \sigma$; otherwise,

  $$|a_{11}||a_{rr}| - (a_{r1})^2 \leqslant |a_{11}|\sigma - \alpha(a_{r1})^2 < 0,$$

a contradiction with the fact (3). Hence, for all i and j, we have that [1]

$$\left|a_{ij}^{(1)}\right| = |b_{ij}^{(1)}| = \begin{cases} |b_{ij}| - \left|\frac{b_{ir}b_{rj}}{b_{rr}}\right| \leqslant |b_{ij}| = |a_{ij}| \leqslant \max\limits_{i,j} |a_{ij}|, \\ -|b_{ij}| + \left|\frac{b_{ir}b_{rj}}{b_{rr}}\right| \leqslant \left|\frac{b_{ir}b_{rj}}{b_{rr}}\right| \leqslant |b_{rj}| = |a_{rj}| \leqslant \max\limits_{i,j} |a_{ij}|. \end{cases}$$

- Case 3. The pivot is the 2×2 submatrix $\begin{bmatrix} a_{11} & a_{1r} \\ a_{r1} & a_{rr} \end{bmatrix}$. Thus, Algorithm 1 makes sure that

$$|a_{11}||a_{rr}| - (a_{r1})^2 \leqslant |a_{11}||a_{rr}| - \alpha^2(a_{r1})^2 < 0,$$

a contradiction with the fact (3). So, the case is excluded.

Notice that S' and S_1S' are signature matrices. Therefore, it has been shown that the result is true for the case $\epsilon_2 = 1$. Now let $\epsilon_2 = -1$. According to the results from [10, pp. 217-218], obviously we have

$$\left|a_{ij}^{(1)}\right| \leqslant (2.57)^s \max_{i,j} |a_{ij}|,$$

where $s = 1$ or 2 is the order of the pivot. Moreover, applying Lemma 1 to the fact (2), we get that $A^{(1)}$ is signature similar to a nonsingular sign regular matrix with signature $\left(\frac{\epsilon_{s+1}}{\epsilon_s}, \frac{\epsilon_{s+2}}{\epsilon_s}, \ldots, \frac{\epsilon_n}{\epsilon_s}\right)$. Thus, the lemma is proved. □

Our main result in this section is the following:

Theorem 1. *Let $A \in \mathbb{R}^{n\times n}$ $(n \geqslant 2)$ be symmetric and signature similar to a nonsingular sign regular matrix with signature $\epsilon \in \mathbb{R}^{1\times n}$, and let us apply the Bunch–Kaufman pivoting strategy to A. Then $\rho_n \leqslant (2.57)^{2\phi(\epsilon)}$.*

Proof. Set $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ and $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$. Let

$$\gamma_1 = \begin{cases} 1, & \text{if } \epsilon_2 = -1, \\ 0, & \text{if } \epsilon_2 = 1, \end{cases} \quad , \quad \gamma_2 = |\{i|\epsilon_i \neq \epsilon_{i+2},\ i = 1, \ldots, n-2\}|.$$

Then we easily see that $\phi(\epsilon) = \gamma_1 + \gamma_2$. Now let us apply the Bunch–Kaufman pivoting strategy to A in the following procedure:

$$A \to A^{(1)} \to \cdots \to A^{(n-1)}, \tag{6}$$

where $A^{(k)} = (a_{ij}^{(k)})$ is the Schur complement during the factorization process. When one step of the Bunch–Kaufman factorization is applied to A, according to Lemma 2, the obtained Schur complement $A^{(1)} = \left(a_{ij}^{(1)}\right)$ satisfies that for all i, j,

$$\left|a_{ij}^{(1)}\right| \leqslant (2.57)^{2\gamma_1} \max_{i,j} |a_{ij}|,$$

[1] Here, $b_{ij}^{(1)}$ should be $b_{i-1,j-1}^{(1)}$ or $b_{i-2,j-2}^{(1)}$. For the convenience of the statement, we abuse the notation.

and $A^{(1)}$ is signature similar to a nonsingular sign regular matrix with signature $\epsilon' = (\epsilon'_i)$ where if the pivot is 1×1, then

$$\epsilon'_1 = \frac{\epsilon_2}{\epsilon_1}, \epsilon'_2 = \frac{\epsilon_3}{\epsilon_1}, \ldots, \epsilon'_{n-1} = \frac{\epsilon_n}{\epsilon_1};$$

if the pivot is 2×2, then

$$\epsilon'_1 = \frac{\epsilon_3}{\epsilon_2}, \epsilon'_2 = \frac{\epsilon_4}{\epsilon_2}, \ldots, \epsilon'_{n-2} = \frac{\epsilon_n}{\epsilon_2}.$$

Obviously, $\phi(\epsilon') \leqslant \gamma_2$. Therefore, we conclude by applying an induction method to the Schur complement $A^{(1)}$ that for all k,

$$\max_{i,j} \left|a_{ij}^{(k)}\right| \leqslant (2.57)^{2\phi(\epsilon')} \max_{i,j} \left|a_{ij}^{(1)}\right| \leqslant (2.57)^{2\phi(\epsilon')+2\gamma_1} \max_{i,j} |a_{ij}| \leqslant (2.57)^{2\phi(\epsilon)} \max_{i,j} |a_{ij}|.$$

Thus, the result is proved. □

Two important subclasses of sign regular matrices are totally nonnegative matrices and totally nonpositive matrices. Clearly, symmetric totally nonnegative matrices are positive definite, but, symmetric totally nonpositive matrices must be indefinite. Set

$$J_n = \mathrm{diag}\{1, -1, \ldots, (-1)^{n-2}, (-1)^{n-1}\} \in \mathbb{R}^{n\times n}.$$

According to [1, Theorem 3.3], if $A \in \mathbb{R}^{n\times n}$ is nonsingular totally nonpositive, then $J_n A^{-1} J_n$ is sign regular with signature $\epsilon = (1, 1, \ldots, 1, -1) \in \mathbb{R}^{1\times n}$, and so, A^{-1} is signature similar to a sign regular matrix. Thus, we have the following result.

Corollary 1. *Let us apply the Bunch–Kaufman pivoting strategy to a symmetric indefinite matrix $A \in \mathbb{R}^{n\times n}$. If A is a nonsingular totally nonpositive matrix, then $\rho_n \leqslant (2.57)^2$; and if A is the inverse of a totally nonpositive matrix, then $\rho_n \leqslant 2.57$.*

Proof. If A is totally nonpositive, then its signature $\epsilon = (-1, \ldots, -1) \in \mathbb{R}^{1\times n}$, and thus, Theorem 1 makes sure that $\rho_n \leqslant (2.57)^2$ because $\phi(\epsilon) = 1$. Now assume that A is the inverse of a totally nonpositive matrix. As mentioned previously, A is signature similar to a sign regular matrix with signature $\epsilon' = (1, \ldots, 1, -1) \in \mathbb{R}^{1\times n}$. So $\phi(\epsilon') = 1$. Although we can get by using Theorem 1 that $\rho_n \leqslant (2.57)^2$, there is something to be said. Let us apply the Bunch–Kaufman pivoting strategy to A in the procedure (6). Then, using Lemma 2, we have that for $1 \leqslant k \leqslant n-2$,

$$\left|a_{ij}^{(k)}\right| \leqslant \max_{i,j} \left|a_{ij}^{(k-1)}\right| \leqslant \cdots \leqslant \max_{i,j} \left|a_{ij}^{(1)}\right| \leqslant \max_{i,j} |a_{ij}|, \quad \text{for all } i, j.$$

Now, if the final pivot in the factorization is 1×1, then according to the results from [10, pp. 217-218], the obtained Schur complement $A^{(n-1)} = \left(a_{ij}^{(n-1)}\right) \in \mathbb{R}^{1\times 1}$ satisfies that

$$|a_{11}^{(n-1)}| \leqslant 2.57 \max_{i,j} \left|a_{ij}^{(n-2)}\right| \leqslant 2.57 \max_{i,j} |a_{ij}|;$$

otherwise, $A^{(n-2)}$ is the pivot, and so there is nothing to do. Therefore, we get that the growth factor ρ_n is bounded by 2.57. The result is proved. □

Remark 1. As shown by Corollary 1, the bound is sometimes an overestimate. Combining flips in the signature with the order of the pivots, smaller bounds of the growth factor can be obtained. Let us use the following example to illustrate the fact. Let $A \in \mathbb{R}^{5\times 5}$ be a nonsingular sign regular matrix with

signature $\epsilon = (1, -1, 1, 1, 1)$. Then $\phi(\epsilon) = 2$. If the first two pivots are 2×2, then $\rho_5 \leqslant 2.57^4$; if the first pivot is 2×2 and other pivots are 1×1, then $\rho_5 \leqslant 2.57^3$; if all the pivots are 1×1, then $\rho_5 \leqslant 2.57^2$.

We easily get the following result for the case that all the pivots are 1×1.

Corollary 2. *Let $A \in \mathbb{R}^{n\times n}$ $(n \geqslant 2)$ be symmetric and signature similar to a nonsingular sign regular matrix with signature $\epsilon \in \mathbb{R}^{1\times n}$, and let us apply the Bunch–Kaufman pivoting strategy to A. If all the pivots are 1×1, then $\rho_n \leqslant (2.57)^{\phi(\epsilon)}$.*

3. The modified Bunch–Kaufman factorization

As pointed out in Introduction section, the Bunch–Kaufman pivoting strategy in general can destroy the band structure of banded matrices. In this section, it is interesting to show that the modified Bunch–Kaufman pivoting strategy proposed by Sorensen and Van Loan [5, Section 5.3.2] can preserve the bandwidth of banded symmetric matrices which are signature similar to sign regular matrices.

A matrix $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is called m-banded if $a_{ij} = 0$ for all $|i - j| > m$. We say that an m-banded matrix $A \in \mathbb{R}^{n\times n}$ is full if $m = n - 1$. The following result presents the signature structure of a nonsingular m-banded sign regular matrix.

Lemma 3 [14]**.** *Let $A \in \mathbb{R}^{n\times n}$ be a nonsingular m-banded $(m < n-1)$ sign regular matrix with signature $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$, then $\epsilon_2 = \epsilon_1^2, \epsilon_3 = \epsilon_1^3, \ldots, \epsilon_{n-m} = \epsilon_1^{n-m}$.*

Next we give the behavior of the matrix structure when one step of the modified Bunch–Kaufman factorization is performed.

Lemma 4. *Let $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ $(n \geqslant 2)$ be symmetric and signature similar to a nonsingular sign regular matrix with signature $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$ where $\epsilon_2 = 1$, and let us apply one step of the modified Bunch–Kaufman factorization to A. Then a row interchange is not necessary to obtain the Schur complement $A^{(1)}$. Furthermore, $A^{(1)} = \left(a_{ij}^{(1)}\right)$ is signature similar to a nonsingular sign regular matrix with signature $\left(\frac{\epsilon_2}{\epsilon_1}, \frac{\epsilon_3}{\epsilon_1}, \ldots, \frac{\epsilon_n}{\epsilon_1}\right)$, and $\left|a_{ij}^{(1)}\right| \leqslant \max\limits_{i,j} |a_{ij}|$ for all i and j.*

Proof. Assume that there exists a signature matrix $S \in \mathbb{R}^{n\times n}$ such that $B = SAS = (b_{ij})$ is nonsingular sign regular with signature $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$ where $\epsilon_2 = 1$. Thus,

$$|a_{11}||a_{jj}| - a_{1j}^2 = |b_{11}||b_{jj}| - b_{1j}^2 = \det B[1, j] \geqslant 0, \quad j \neq 1,$$

from which it follows that

$$|a_{11}|\sigma' \geqslant |a_{11}||a_{jj}| \geqslant a_{1j}^2 \geqslant \alpha a_{1j}^2, \quad \text{where } \sigma' = \max_i |a_{ij}|, \ \forall j \neq 1,$$

and so, the modified Bunch–Kaufman pivoting strategy guarantees that a_{11} must be the pivot. This means that no interchange is done in order to obtain the Schur complement $A^{(1)} = \left(a_{ij}^{(1)}\right)$. Notice that $\epsilon_2 = 1$. Thus, according to the proof of Lemma 2, it is easy to get that $A^{(1)}$ is signature similar to a nonsingular sign regular matrix with signature $\left(\frac{\epsilon_2}{\epsilon_1}, \frac{\epsilon_3}{\epsilon_1}, \ldots, \frac{\epsilon_n}{\epsilon_1}\right)$, and $\left|a_{ij}^{(1)}\right| \leqslant \max\limits_{i,j} |a_{ij}|$ for all i and j. The result is proved. □

Our main result in this section is the following:

Theorem 2. *Let $A \in \mathbb{R}^{n\times n}$ $(n \geqslant 2)$ be an m-banded symmetric matrix which is signature similar to a nonsingular sign regular matrix with signature $\epsilon \in \mathbb{R}^{1\times n}$, and let us apply the modified Bunch–Kaufman pivoting strategy to A. Then the band structure is preserved in the whole factorization process with the growth factor $\rho_n \leqslant (2.57)^{2\phi(\epsilon)}$.*

Proof. Let us apply the modified Bunch–Kaufman pivoting strategy to $A \in \mathbb{R}^{n\times n}$ in the procedure (6). If $m = n - 1$, then it is trivial that the band structure is preserved because A is full. Now assume that $m < n - 1$. From Lemma 3, the m-banded symmetric matrix $A = (a_{ij})$ is signature similar to a nonsingular sign regular matrix with signature $\epsilon = (\epsilon_i) \in \mathbb{R}^{1\times n}$ where

$$\epsilon_2 = \epsilon_1^2, \epsilon_3 = \epsilon_1^3, \ldots, \epsilon_{n-m} = \epsilon_1^{n-m}.$$

So, $\epsilon_2 = 1$. Thus, when one step of the modified Bunch–Kaufman factorization is applied to A, Lemma 4 guarantees that there is no interchange to be done such that the obtained Schur complement $A^{(1)} = \left(a_{ij}^{(1)}\right) \in \mathbb{R}^{(n-1)\times(n-1)}$ is an m-banded symmetric matrix, which is signature similar to a nonsingular sign regular matrix with signature $\epsilon' = \left(\epsilon_i'\right) \in \mathbb{R}^{1\times(n-1)}$ where

$$\epsilon_1' = \frac{\epsilon_2}{\epsilon_1} = \epsilon_1, \epsilon_2' = \frac{\epsilon_3}{\epsilon_1} = \epsilon_1^2, \ldots, \epsilon_{n-m-1}' = \frac{\epsilon_{n-m}}{\epsilon_1} = \epsilon_1^{n-m-1}.$$

Therefore, we conclude by applying an induction method to the Schur complement $A^{(1)}$ that the band structure of A is preserved in the whole factorization process. In addition, using a similar argument as that of Theorem 1, we have that $\rho_n \leqslant (2.57)^{2\phi(\epsilon)}$. The result is proved. □

We easily derive from Lemma 3 that a nonsingular totally nonpositive matrix must be full, and so there is nothing to be said about the preservation of the band structure. According to the proof of Theorem 2, we have the following result for the banded inverse of a totally nonpositive matrix by considering the signature $(1, \ldots, 1, -1)$.

Corollary 3. *Let $A \in \mathbb{R}^{n\times n}$ be a banded symmetric matrix whose inverse is totally nonpositive, and let us apply the modified Bunch–Kaufman pivoting strategy to A. Then at most one interchange is done, and it is done in the last step of the whole factorization process which guarantees the band structure is preserved.*

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